

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1510

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism

Code of Conduct

I K. Sai Pritya (192421296) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Sai Pritya



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Title: Microservices Architecture in Cloud Environments

Prepared by: K. Saipriya Register Number: 192421296

OBJECTIVE

- To understand the concept of Microservices Architecture.
- To learn how microservices work in cloud environments.
- To explore its architecture and key components.
- To understand its industry relevance and practical applications.

What is Microservices Architecture?

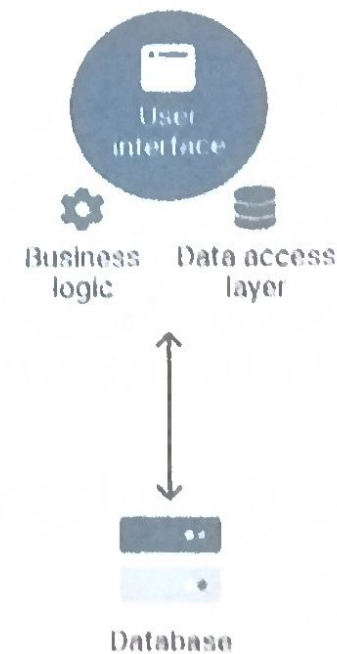
Definition

- A software architecture that divides an application into small, independent services.
- Each service performs a specific business function.
- Services communicate using APIs.

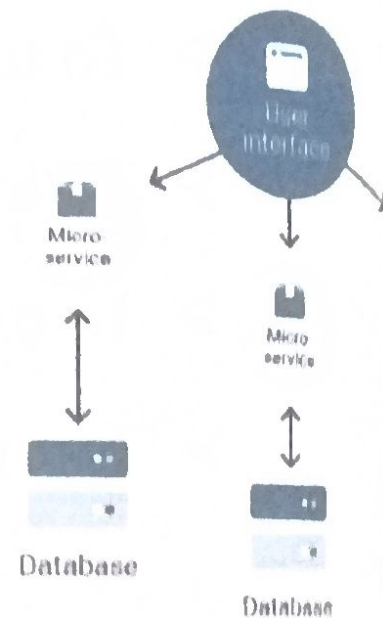
Key Characteristics

- Independent services
- Loosely coupled
- API-based communication
- Independent deployment
- High scalability

Monolithic Architecture



Microservice Architect



Core Technical Explanation

How Microservices Work

Key Components

- **Client/User** – Sends requests to the application.
- **API Gateway** – Receives and routes requests.
- **Microservices** – Independent services handling specific tasks.
- **Database** – Each service has its own database.
- **APIs** – Enable communication between services.

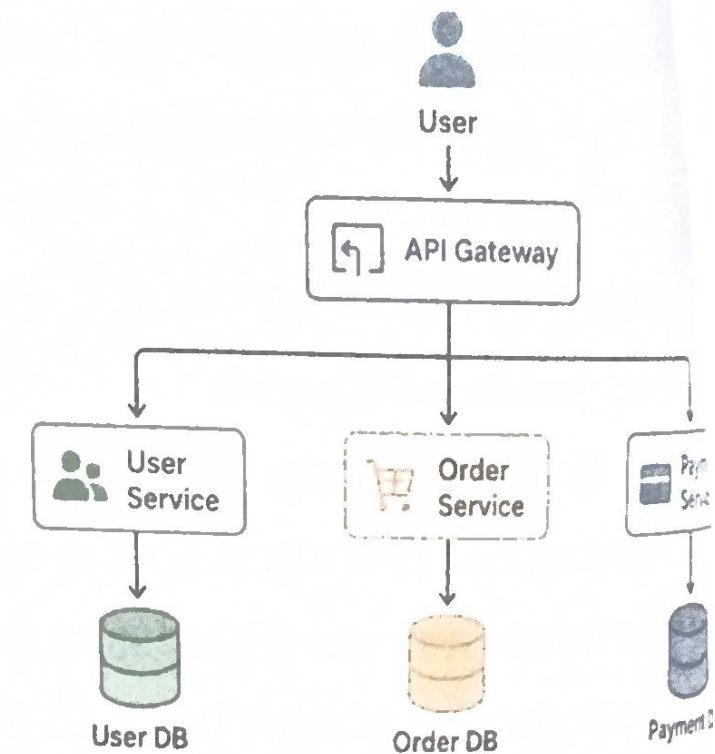
Working Process

- User sends a request.
- API Gateway receives the request.
- The request is routed to the appropriate microservice.
- The microservice processes the request and accesses its database.
- The response is sent back to the user.

Microservices Architecture

Architecture Overview

- Client sends requests through the API Gateway.
- API Gateway routes requests to the appropriate microservice.
- Each microservice performs a specific business function.
- Every microservice has its own dedicated database.
- Services communicate using APIs.



Industry Relevance of Microservices

Industries Using Microservices

- **E-commerce** – Amazon
- **Video Streaming** – Netflix
- **Ride Sharing** – Uber
- **Music Streaming** – Spotify
- **Online Payments** – PayPal

The Amazon logo, featuring the word "amazon" in a bold, lowercase, sans-serif font. A curved orange arrow is positioned below the letters, starting under the 'a' and ending under the 'z'.

Why Industries Prefer Microservices

- Handles millions of users efficiently
- Easy to scale applications
- Faster software updates
- High availability and reliability
- Reduced downtime during deployments

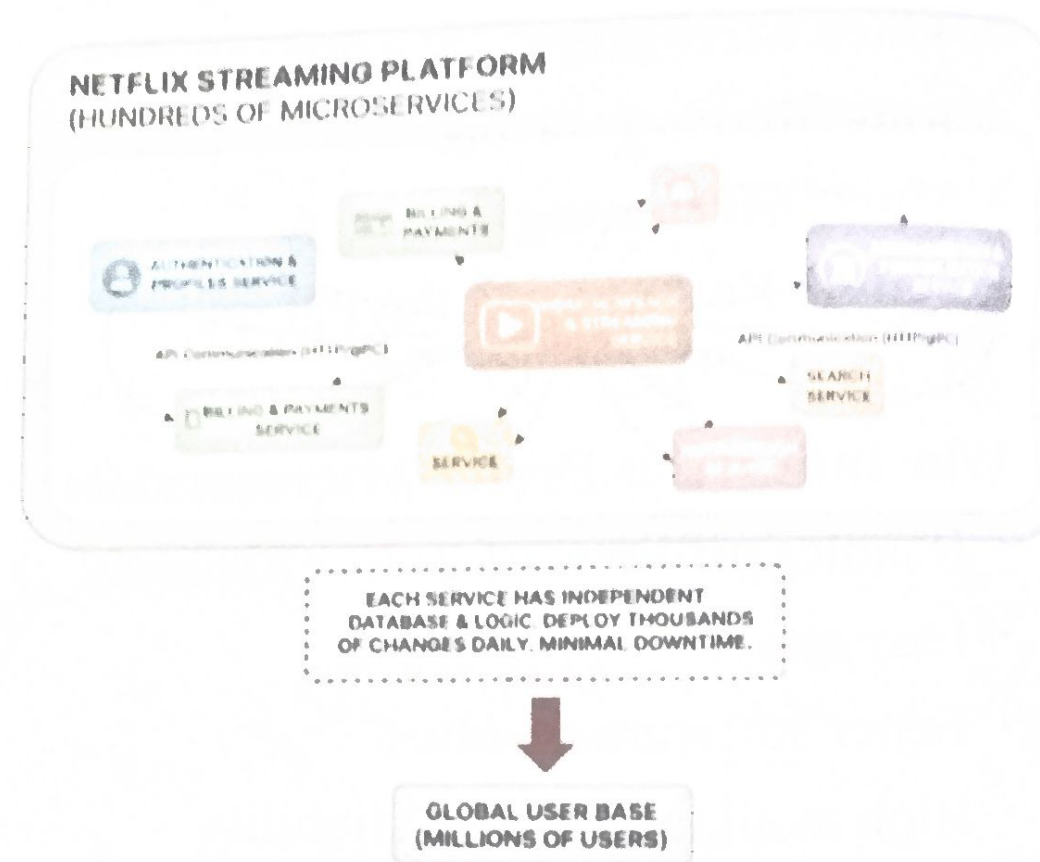
The Netflix logo, consisting of the word "NETFLIX" in a bold, uppercase, sans-serif font. The letters are a dark red color.

Real-World Use Case: Netflix

Netflix and Microservices

- Uses microservices for different functions.
- Authentication Service – User login
- Recommendation Service – Suggests movies
- Streaming Service – Plays videos
- Payment Service – Manages subscriptions
- Notification Service – Sends updates
- Supports millions of users efficiently

NETFLIX MICROSERVICE ARCHITECTURE



Advantages & Challenges

Advantages

Easy to scale

Independent deployment

Faster development

Better fault isolation

High availability

Challenges

Complex management

Network latency

Security concerns

Data consistency

Monitoring multiple services

Conclusion & Key Takeaways

Conclusion

Microservices Architecture enables organizations to build scalable, flexible, and reliable cloud applications by dividing them into independent services.

Key Takeaways

- Applications are divided into small, independent services.
- Services communicate using APIs.
- Easy to develop, deploy, and scale.
- Widely used by companies like Netflix, Amazon, and Uber.
- A key architecture for modern cloud computing.